

Software requirement Specification

1. Introduction

1.1 Purpose

The Software Requirements Specification is to define the requirements for the Smart Parking Platform that will be developed by WE ARE \$oftware corp. It will serve as a guide throughout the planning and development of the system.

1.2 Product Overview

The Smart Parking Platform is a software system that will be available through the web or on a mobile device. The Smart Parking Platform will help drivers locate, reserve and pay for parking garages in real time. The system will provide users with parking availability information, navigation assistance and management tools for parking operators.

2 Competitive Landscape and Market Research

2.1 Market Overview

The smart parking market focuses on using technology to make finding, reserving, paying, and managing parking more efficient.

2.2 Existing Solutions

There are several parking platforms that provide drivers with tools for finding, reserving, and paying for parking. Three popular parking applications are ParkWhiz and SpotHero.

SpotHero is available through the web and mobile applications. Drivers are able to search for parking near a destination, compare locations and prices, reserve a space and pay in advance. SpotHero offers hourly, monthly, airport, and event parking across 400 cities in the US and Canada. One limitation is that SpotHero does not own or operate the parking facilities listed on its platform and relies on parking partners to provide inventory information and availability.

ParkWhiz allows users to search for parking by address, venue, or other destination through its website and mobile application. Users can compare nearby parking locations and prices, select a time period, reserve a space, and pay digitally. A strength ParkWhiz has is the variety of payment options they offer and their parking options. However, parking rates and available services can vary depending on the individual parking provider.

2.3 Market Gaps and Product Differentiation

The existing parking applications provide many useful services, however there are still opportunities to create a more integrated parking experience for drivers and parking operators.

- Individual Space Availability: Drivers can see specific available parking spaces within garages rather than just seeing its availability.
- Indoor Parking Navigation: Drivers can receive directions to their available or reserved space after entering a garage.
- Space-Level Monitoring: Parking operators can monitor the availability and occupancy of individual spaces in real time

3 Vision and Scope

3.1 Vision

The vision is to build an efficient easy to use parking application that reduces the time and frustration associated with finding available parking. The system will connect drivers with real-time parking information while giving parking operators the tools needed to better monitor and manage their facilities.

3.2 Objectives

- Reduce the amount of time drivers spend searching for available parking spaces.
- Provide accurate real-time parking availability information.
- Allow users to reserve and pay for parking through the application.
- Give parking operators tools to monitor occupancy and manage their parking facilities
- Provide occupancy report and analytics to help operators improve the utilization of available parking.
- Make parking a convenient and secure experience for both drivers and operators

3.3 Scope

The system will be available through mobile applications and the web and will support both drivers and parking facility operators

4 User Stories

1. As a driver, I want to create an account so I can access the parking services.
2. As a driver, I want to log into my account securely so I can access my personal information.
3. As a driver, I want indoor navigation to my available or reserved space so I can quickly and easily find the space after entering the garage.
4. As a driver I want to pay for parking digitally so that I can complete my payment through the application.
5. As a driver, I want to view parking spots on an interactive indoor map so I can easily locate spaces within the parking garage.

6. As a driver, I want to reserve a spot so I can ensure a space will be available when I arrive.
7. As a driver, I want to view my reservation history and receipts so I can track my previous transactions.
8. As a driver, I want to receive notifications and alerts so I can be aware of any updates on my parking reservations.
9. As a driver I want to have navigation assistance so that I can easily find the parking facility.
10. As a parking operator, I want an administrative dashboard so that I can manage my parking facility.
11. As a parking operator, I want to view real-time occupancy reports and analytics so I can understand activity and utilization.
12. As a parking operator, I want to monitor occupancy in real time so I know how much parking is available.
13. As a driver, I want to view real-time parking availability so that I can find parking more easily.
14. As a driver, I want to see which individual parking spaces are available in real time so that I can locate an open space before entering a garage.
15. As a parking operator, I want to monitor the occupancy of individual parking spaces so that I can see exactly which spaces are available or occupied.